

Microsoft Corporation (MSFT)

Is AI Capital Spending Creating or Destroying Value?

A Simplified DCF and Capex-Efficiency Analysis

Equity Research Practice Report | Prepared August 18, 2026 | Sector: Technology / Cloud Infrastructure

Rating HOLD	Price (8/18/26) \$500	Implied Value \$337
Market Cap \$3.7T	Trailing P/E 27.6x	Base-Case Downside (32.5%)

1 Investment Thesis

Microsoft's fiscal 2026 results crystallize the central question facing every hyperscaler: capital expenditure on AI infrastructure is now growing far faster than the revenue it is meant to produce, and the market is being asked to trust that the gap closes. In FY2026, Microsoft's capital expenditures (property and equipment, cash basis) reached \$115.95 billion, up 79.6% year-over-year, while total company revenue grew 17.8% to \$331.84 billion. That divergence has *accelerated* within the year: capex plus finance-lease additions grew 49% year-over-year in fiscal Q3 and 69% year-over-year in fiscal Q4.

This report builds a simplified unlevered discounted cash flow (DCF) model seeded with Microsoft's actual FY2026 results and tests what has to be true — in growth, margin, and capital-intensity terms — for the current \$500 share price (\$3.7 trillion market capitalization) to be justified. Under a base case that assumes capex intensity normalizes gradually from its current elevated level and operating margin recovers as that normalization occurs, the model's discounted cash flow implies a value of approximately **\$337 per share**, roughly 32% below the current price. Even a constructive bull case in which margins expand faster and capex normalizes sooner implies a value of approximately \$399, still below market. Closing that gap requires either a lower cost of capital than this report assumes (WACC near 8.0% rather than 9.0%) or sustained terminal growth above 4.0% in perpetuity — both aggressive assumptions for a company already generating \$331.84 billion in revenue.

The conclusion is not that Microsoft's AI investment is unsound. Azure revenue growth accelerated to 43% (constant currency) in fiscal Q4, commercial remaining performance obligations rose 84% year-over-year to \$678 billion (more than twice annual revenue), and management indicated calendar-2026 capex guidance was in fact revised *down* (to approximately \$175 billion from \$190 billion) by extending the assumed useful life of data center and office assets from 15 to 25 years — an accounting change that itself flatters near-term margins. The thesis here is narrower: **the current valuation already prices in a highly favorable resolution of the capex-to-revenue gap**, leaving limited room for execution missteps, further capex surprises, or a slower-than-expected monetization curve. We rate the stock **HOLD** pending clearer evidence that revenue growth is reaccelerating in line with capex growth, rather than merely tracking a stable 18% pace.

2 The Core Tension: Capex Growth vs. Revenue Growth

Figure 1 shows the widening gap between Microsoft's revenue growth, which has held in a narrow 15–18% band across the last five reported periods, and capital expenditure growth, which has climbed from 49% to

69% year-over-year within fiscal 2026 alone.

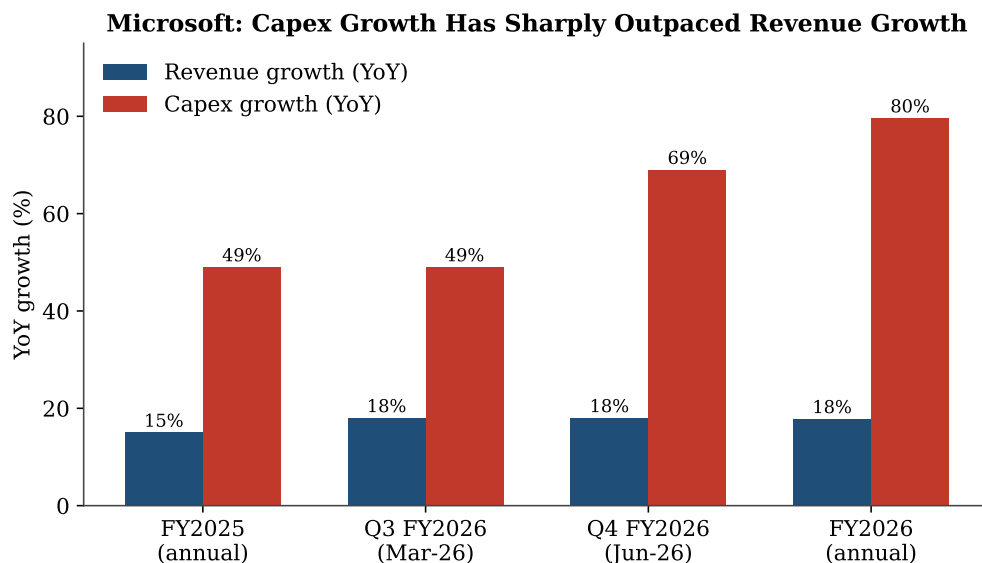


Figure 1: Microsoft revenue growth vs. capital expenditure growth, FY2025–FY2026. Capex figures are cash purchases of property and equipment (comparable basis); Q3/Q4 FY2026 also disclosed capex plus finance leases growing 49% and 69% respectively. *Source: Microsoft 10-K (FY2025), Q3 and Q4 FY2026 earnings releases and calls.*

This is not unique to Microsoft. Amazon’s Q2 2026 AWS segment posted 36.7% revenue growth against company-wide cash capex of \$53.1 billion for the quarter (up from \$32.1 billion a year earlier), and management raised full-year 2026 capex guidance twice in six months, from roughly \$200 billion to \$220 billion. Section 4 places the two companies side by side.

The bull case for the sector rests on three observable, non-speculative facts: backlogs are growing faster than capex (Microsoft’s RPO grew 84% against capex growth of 70–80%), operating margins on the cloud businesses that management does disclose are expanding (AWS margin rose from 32.9% to 39.4% year-over-year even amid the spending surge), and useful-life assumptions on the underlying assets are lengthening, which will mechanically reduce depreciation drag in coming years. The bear case is that revenue growth has not yet shown the same order-of-magnitude acceleration as capex growth, and a discounted cash flow built on today’s numbers does not yet clear the current share price without assumptions more optimistic than the base case supports.

3 Valuation: Simplified DCF

3.1 Methodology and Key Assumptions

The model projects unlevered free cash flow for FY2027E–FY2031E off a FY2026A base of \$331.84 billion in revenue and \$155.0 billion in operating income (both as reported). Revenue growth decelerates from 15% to 7% across the forecast window; operating margin is assumed to dip slightly in FY2027 (44%, versus 46.7% actually realized in FY2026) before recovering as elevated capex normalizes, reaching 47% by FY2031. Capex intensity, at 34.9% of revenue in FY2026A, is assumed to decline toward a 20% long-run norm by FY2031 — still above Microsoft’s pre-AI historical range of roughly 15%, reflecting a structurally higher-capex cloud/AI business mix. Depreciation and amortization is modeled at 55% of same-year capex, a simplification for the lagged effect of the useful-life extension disclosed in the Q4 FY2026 call. All pro-

jected cash flows are discounted at a 9.0% WACC (consistent with a low-beta, investment-grade mega-cap) and terminal value is calculated using Gordon Growth at a 3.5% terminal growth rate. Cash, investments, and debt are taken directly from the FY2026 Q4 balance sheet.

(\$ millions)	FY26A	FY27E	FY28E	FY29E	FY30E	FY31E
Revenue growth %	—	15.0%	13.0%	11.0%	9.0%	7.0%
Revenue	331,840	381,616	431,226	478,661	521,740	558,262
Operating margin %	46.7%	44.0%	45.0%	46.0%	46.5%	47.0%
Operating income (EBIT)	155,000	167,911	194,052	220,184	242,609	262,383
NOPAT (19% tax)	125,550	136,008	157,182	178,349	196,514	212,530
Capex % of revenue	34.9%	30.0%	26.0%	23.0%	21.0%	20.0%
Capex	(115,950)	(114,485)	(112,119)	(110,092)	(109,565)	(111,652)
D&A (addback)	63,772	62,967	61,665	60,551	60,261	61,409
Unlevered FCF	73,372	84,490	106,728	128,808	147,209	162,287

Table 1: Simplified unlevered FCF build. FY26A figures are actual reported results; FY27E–FY31E are this model’s projections. See Section 6 for line-item sourcing.

3.2 DCF Output

Item	Value
Sum of PV of explicit FCF (FY27E–FY31E)	\$476,570mm
Terminal value (undiscounted, end of FY31E)	\$3,053,943mm
PV of terminal value	\$1,984,854mm
Enterprise Value	\$2,461,423mm
Plus: Cash & short-term investments	\$76,800mm
Less: Total debt	(\$31,100mm)
Equity Value	\$2,507,123mm
Shares outstanding	7,430mm
Implied share price	\$337.43
Current share price (8/18/26)	\$500.00
Implied upside / (downside)	(32.5%)

Table 2: DCF valuation summary. 80.6% of enterprise value derives from the terminal value, underscoring how sensitive this output is to long-run growth and margin assumptions well beyond the explicit forecast window — the central caveat on any mega-cap DCF.

3.3 Sensitivity

Figure 2 stress-tests the two assumptions the valuation is most sensitive to: discount rate and terminal growth. Only in the most optimistic corner of this grid — an 8.0% WACC paired with 4.5% perpetual growth — does the base-case operating model clear the current \$500 share price. Every other combination implies the stock is priced for a more favorable outcome than this model’s base case delivers.

3.4 Scenario Summary

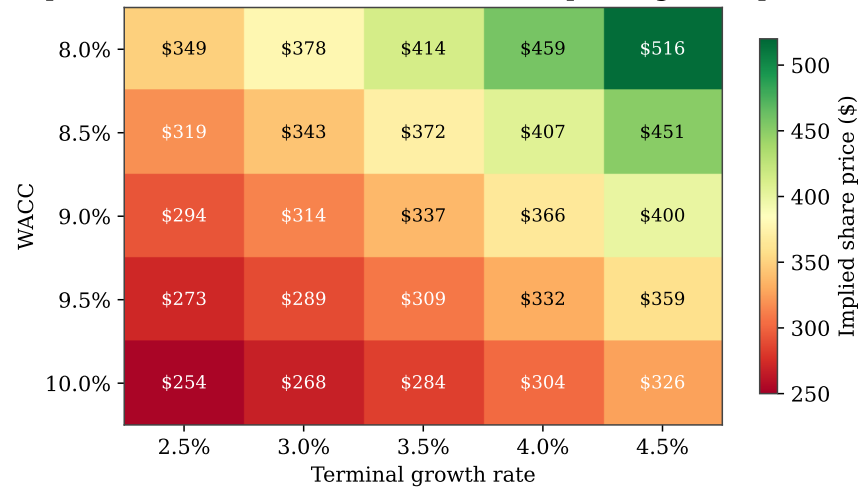
Implied MSFT Share Price (\$) — Base-Case Operating Assumptions

Figure 2: Implied MSFT share price across a grid of WACC and terminal growth assumptions, holding operating margin and capex-intensity assumptions at base case. Green shading indicates values at or above the current \$500 share price.

Scenario	Implied Price	Enterprise Value	vs. Current Price
Bear (margins stall near 43%, capex stays elevated)	\$283.82	\$2,063,056mm	(43.2%)
Base case	\$337.43	\$2,461,423mm	(32.5%)
Bull (margins reach 51%, capex normalizes to 15% by FY31)	\$399.32	\$2,921,244mm	(20.1%)

Table 3: All three scenarios held at a 9.0% WACC and 3.5% terminal growth rate for comparability; only operating margin and capex-intensity trajectories vary. Even the bull case does not clear the current share price at this discount rate.

4 Comparable Benchmark: Microsoft vs. Amazon (AWS)

Because Microsoft does not disclose Azure's standalone operating margin, Amazon's more granular AWS segment reporting provides the closest read-through on whether hyperscale cloud/AI capex is converting into segment profitability.

Two things stand out. First, Amazon's willingness to disclose AWS-specific operating margin — and to show it expanding even as capex accelerates — is a data point Microsoft does not offer investors for Azure; the market is effectively asked to trust Microsoft's aggregate operating margin trend as a proxy. Second, Microsoft's capex burden relative to its own revenue base (45.6% of quarterly revenue) is nearly double Amazon's (26.5%), because Amazon's capex is diluted by a much larger e-commerce revenue base that AWS is not required to support directly. This makes Microsoft's aggregate operating margin more sensitive to a stall in cloud/AI monetization than Amazon's.

5 Key Risks to the Thesis

1. **Revenue reacceleration.** If Azure growth continues climbing from 43% toward the 50%+ range implied by the \$678 billion RPO working through the P&L, base-case revenue growth assumptions in this model (15% decelerating to 7%) would prove too conservative, closing much of the valuation gap on their own.

Metric (most recent quarter)	Microsoft	Amazon (AWS)
Cloud segment revenue	\$39.31bn	\$42.20bn
Cloud revenue growth YoY	32%	36.7%
Cloud constant-currency growth	43% (Azure)	n/a
Cloud operating margin	Not disclosed	39.4% (up from 32.9%)
Company-wide quarterly capex	\$41.0bn	\$53.1bn
Capex growth YoY	69%	~65–70% (TTM)
FY2026 capex guidance	~\$175bn (revised down)	~\$220bn (revised up)
Contracted backlog / RPO	\$678bn (+84% YoY)	\$496bn (triple-digit growth)
Capex as % of company revenue	45.6%	26.5%
Market capitalization	~\$3.7tn	~\$2.55tn

Table 4: Source: Microsoft Q4 FY2026 earnings release/call (July 29, 2026); Amazon Q2 2026 earnings release/call (July 30, 2026). Capex-as-%-of-revenue uses each company's total quarterly revenue (\$90.01bn MSFT; \$200.6bn AMZN) as the denominator.

2. **Useful-life accounting change.** Extending depreciable life on data centers from 15 to 25 years mechanically reduces near-term D&A and lifts reported operating margin — a legitimate accounting judgment, but one that makes reported margin trends harder to compare against history.
3. **Capacity-constrained upside.** Management has repeatedly said Azure demand exceeds supply; if that persists, capex is arguably not optional spend but backlog Microsoft cannot yet fulfill, which would support a more bullish revenue trajectory than modeled here.
4. **Financing and rate risk.** A DCF at this scale is highly sensitive to WACC (Figure 2). A sustained rise in long-term rates, or investor demand for a higher equity risk premium on AI capex risk specifically, would widen the model's implied downside further.
5. **Model simplification risk.** This is a company-level, not segment-level, DCF; a full research build would separately value Azure, Microsoft 365/Productivity, and More Personal Computing, each with different growth and margin profiles, and would model working capital and stock-based compensation dilution explicitly rather than assuming them away.

6 Sources

- Microsoft Corp., Q4 FY2026 earnings release and conference call transcript, July 29, 2026.
- Microsoft Corp., FY2025 Form 10-K and FY2025 Annual Report (SEC EDGAR).
- Microsoft Corp., Q1 and Q3 FY2026 earnings releases (SEC EDGAR, Form 8-K exhibits).
- Amazon.com Inc., Q2 2026 earnings release and conference call transcript, July 30, 2026.
- Market data (share price, market capitalization, P/E, EV/EBITDA, ROIC): FinanceCharts, GuruFocus, StockAnalysis.com, and CompaniesMarketCap, as of the week of August 11–18, 2026.

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